

Customer Churn Analysis & Prediction Report

1. Executive Summary

Customer churn is a critical business problem that directly impacts revenue and growth. This project analyzes customer data to identify churn patterns, uncover key drivers, and build a predictive model to identify customers at risk of leaving.

The analysis reveals that **contract type, pricing, and service quality** are the primary drivers of churn. A machine learning model was also developed to proactively identify high-risk customers with **85% accuracy**.

2. Business Problem

The company is experiencing:

- High customer churn rates
- Lack of visibility into churn drivers
- Inefficient retention strategies

Objective:

- Identify key factors influencing churn
 - Segment high-risk customers
 - Provide actionable business recommendations
 - Predict future churners
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3. Dataset Overview

The dataset includes:

- Customer demographics (Age, Gender, Marital Status)
 - Account information (Tenure, Contract Type, Payment Method)
 - Service usage (Internet Type, Add-on Services)
 - Financial metrics (Monthly Charges, Total Revenue)
 - Churn status, category, and reason
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4. Data Cleaning & Preparation

Performed using **MySQL**:

- Handled missing/null values
- Standardized categorical variables
- Created derived columns (Tenure groups, Age groups)
- Removed inconsistencies and duplicates

SQL Views Created:

- Churned Customers
- Active Customers
- New Joiners

These views improved query efficiency and simplified analysis.

5. Exploratory Data Analysis (EDA)

Key Metrics:

- Total Customers: **6,418**
- Total Churn: **1,732**
- Churn Rate: **27.0%**

Key Findings:

- **Month-to-Month contracts** → 46.5% churn (highest risk)
- **Mailed Check users** → 37.8% churn
- **Fiber Optic users** → 41.1% churn
- **Female customers** contribute ~64% of churn
- Highest churn region: **Jammu & Kashmir (57.2%)**

Behavioral Insights:

- Long-term customers (>24 months) still show churn risk
 - Customers with poor service experience are highly likely to churn
 - Competitor offerings significantly influence churn decisions
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6. Dashboard & Visualization (Power BI)

An interactive dashboard was built to monitor:

- Customer distribution
- Churn trends across segments
- Geographic churn patterns
- Service usage impact

Features:

- KPI cards (Customers, Churn, Rate)
 - Dynamic filters (Age, Monthly Charges, Marital Status)
 - Drill-down capability for deeper analysis
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7. Machine Learning Model

Model Used:

Random Forest Classifier

Performance:

- Accuracy: **85%**
- Precision: 0.83
- Recall: 0.65
- F1 Score: 0.73

Key Predictors:

- Contract Type
- Total Revenue
- Monthly Charges
- Tenure
- Age

The model effectively identifies high-risk customers but can be further improved in recall.

8. Business Recommendations

Based on analysis:

1. **Promote Long-Term Contracts**
 - Offer discounts for yearly subscriptions
 2. **Improve Customer Support**
 - Address service quality issues
 3. **Optimize Pricing Strategy**
 - Re-evaluate fiber pricing
 4. **Target High-Risk Regions**
 - Focus retention campaigns in high churn states
 5. **Encourage Digital Payments**
 - Reduce churn linked to manual payment methods
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9. Conclusion

This project demonstrates a complete data analytics lifecycle:

- Data cleaning and transformation
- Exploratory analysis
- Business intelligence dashboard
- Predictive modeling

The insights generated can help businesses reduce churn, improve customer retention, and make data-driven decisions.

10. Future Improvements

- Improve model recall using advanced techniques (XGBoost, SMOTE)
- Deploy model using APIs for real-time prediction
- Integrate customer feedback data for deeper insights